

Business Requirements – TaxTrace

Client: Meridian Commerce Group **Engagement:** Tax Data Automation, indirect tax validation and reconciliation **Prepared by:** Tax Transformation and Technology **Status:** Baseline, v1.0

1. Client background

Meridian Commerce Group is a mid-market retail and e-commerce group operating across Australia, New Zealand, the United Kingdom and eight United States states. Annual revenue is approximately AUD 480 million.

Growth by acquisition has left Meridian with four finance systems that were never consolidated.

Source system	Region	Approx. share of transactions
SAP ECC	Australia	36%
Microsoft Dynamics 365	United Kingdom	28%
Oracle Fusion	United States	24%
NetSuite	New Zealand	12%

Each system holds the same concepts under different field names, different boolean conventions, and different date formats. Tax rates are configured independently in each, by different people, at different times.

2. The problem

Indirect tax returns are prepared monthly in Excel. A senior analyst downloads extracts from each ERP, joins them by hand, eyeballs the totals against last month, and files. The process takes nine working days.

Three incidents in the last eighteen months have driven this engagement.

- A UK VAT rate change was applied in Dynamics two weeks late. 4,100

transactions were filed at the wrong rate. The error was found by the external auditor, not by Meridian.

- An expired US resale exemption certificate continued to be honoured for seven months, understating collected sales tax.

- A duplicate invoice feed from NetSuite double counted NZD 210,000 of output tax. It was found only because a customer queried their statement.

None of these were found by a control. All three were found by an outsider.

3. Business objective

Automatically determine, validate and reconcile indirect tax on every transaction before filing, so that errors are found by Meridian's own controls rather than by an auditor or a customer, and so that every calculated figure can be traced back to the source record and the rule that produced it.

4. Users

User	What they need
Indirect Tax Manager	Confidence the filing is right, and evidence for why
Tax Analyst	To change a rate or a treatment without a developer
Finance Operations	A clear, categorised exception list to work through
Data Engineering	Reproducible, testable, version controlled transformations
Internal Audit	A traceable line from filed figure back to source row and rule
External Auditor	Documented rules, documented controls, evidence of testing

5. Functional requirements

ID	Requirement	Priority
FR-01	Ingest transaction extracts from all four source systems	Must
FR-02	Map each source schema onto a single canonical transaction model	Must
FR-03	Profile incoming data and report quality before processing	Must
FR-04	Standardise country, currency, region, category and date values	Must
FR-05	Determine the applicable tax code from jurisdiction, product category, customer status and transaction date	Must
FR-06	Apply the rate effective on the transaction date , not today's rate	Must
FR-07	Support tax-inclusive and tax-exclusive source amounts	Must
FR-08	Honour customer exemption certificates only within their validity dates	Must
FR-09	Allow a Tax Analyst to add or change a rule in configuration, with no code change	Must
FR-10	Record rule id and rule version against every calculated result	Must
FR-11	Reconcile calculated tax against source tax and classify each variance	Must
FR-12	Run a named, versioned set of data quality checks with severities	Must
FR-13	Produce an exception report categorised by cause and owner	Must

ID	Requirement	Priority
FR-14	Compare output against a golden dataset and report pass, fail or warning	Must
FR-15	Detect statistical anomalies in tax rate, tax amount and volume	Should
FR-16	Detect duplicate transactions and duplicate invoices	Must
FR-17	Expose results through SQL for ad hoc investigation	Must
FR-18	Log every run with row counts, rules applied, and elapsed time	Must
FR-19	Provide a dashboard summarising the current filing position	Could
FR-20	Evaluate AI generated rule proposals before a human approves them	Could

6. Non-functional requirements

ID	Requirement	Target
NFR-01	Full pipeline run on a month of data	Under 5 minutes on a laptop
NFR-02	Deterministic. Same input and config produce identical output	Exact
NFR-03	Idempotent. Re-running does not duplicate or corrupt results	Exact
NFR-04	Automated test coverage of tax calculation and rule resolution	90%+
NFR-05	Every rule change is version controlled and reviewable	100%
NFR-06	No personally identifying data in logs	100%
NFR-07	Runs locally with free and open source tooling only	Required

7. Assumptions

1. Source extracts are delivered daily as CSV. Real time is out of scope.
2. Jurisdiction is determined by the customer's billing address. Meridian

accepts this simplification for the pilot.

1. United States sales tax is modelled at state level only. County and city

precision is out of scope for the pilot and is a known limitation.

1. Currency conversion uses a monthly average rate, supplied by Finance.
2. Meridian's Tax team owns the rule configuration. Engineering owns the engine.
3. Historical filings will not be restated as part of this engagement.

8. Risks

Risk	Impact	Mitigation
Rule configuration becomes as complex as code	High	Decision table format, mandatory review, versioning
Tax SME and engineer interpret a rule differently	High	Golden dataset agreed and signed off before build
Source system changes a field without notice	Medium	Schema validation on ingest, fail loudly
Exception list too large to action	Medium	Severity and materiality thresholds, ranked by value
Over-reliance on AI generated rules	High	Deterministic validation and mandatory human approval
State level US modelling misses local rates	Medium	Documented limitation, flagged on every US output

9. Acceptance criteria

The pilot is accepted when all of the following hold on one full month.

1. Every transaction in the source extracts appears exactly once in the canonical model, or on the rejection report with a reason.

1. 100% of golden dataset transactions produce the expected tax code, rate and tax amount.

1. Reconciliation classifies 100% of variances into a named category. No variance is left as UNKNOWN.

1. All twelve mandatory data quality checks execute and report.
2. Every calculated row carries a rule id, rule version and timestamp.
3. A Tax Analyst can change a rate in configuration and see the effect, with no

Python change, and the regression suite proves nothing else moved.

1. The pipeline reruns end to end and produces byte-identical output.
2. An auditor can select any filed figure and trace it to the source row and

the rule that produced it, in under two minutes.

Open questions for the Tax SME

Deliberately unanswered. You will resolve these in the Day 3 and Day 7 requirements workshop simulations.

1. When a customer's exemption certificate expires mid-month, is the whole month taxable or only transactions after the expiry date?

1. For a UK sale shipped to Northern Ireland, which rules apply?
2. If a credit note reverses a transaction taxed at an old rate, does the credit use the original rate or the current one?

1. What is the materiality threshold below which a variance is not worth investigating?

1. Freight is currently treated as its own category. Should it follow the tax treatment of the goods it delivers?